High resolution emissions inventory for Megacity of São Paulo

MetroHealth - Instituo de Estudos Avançados USP

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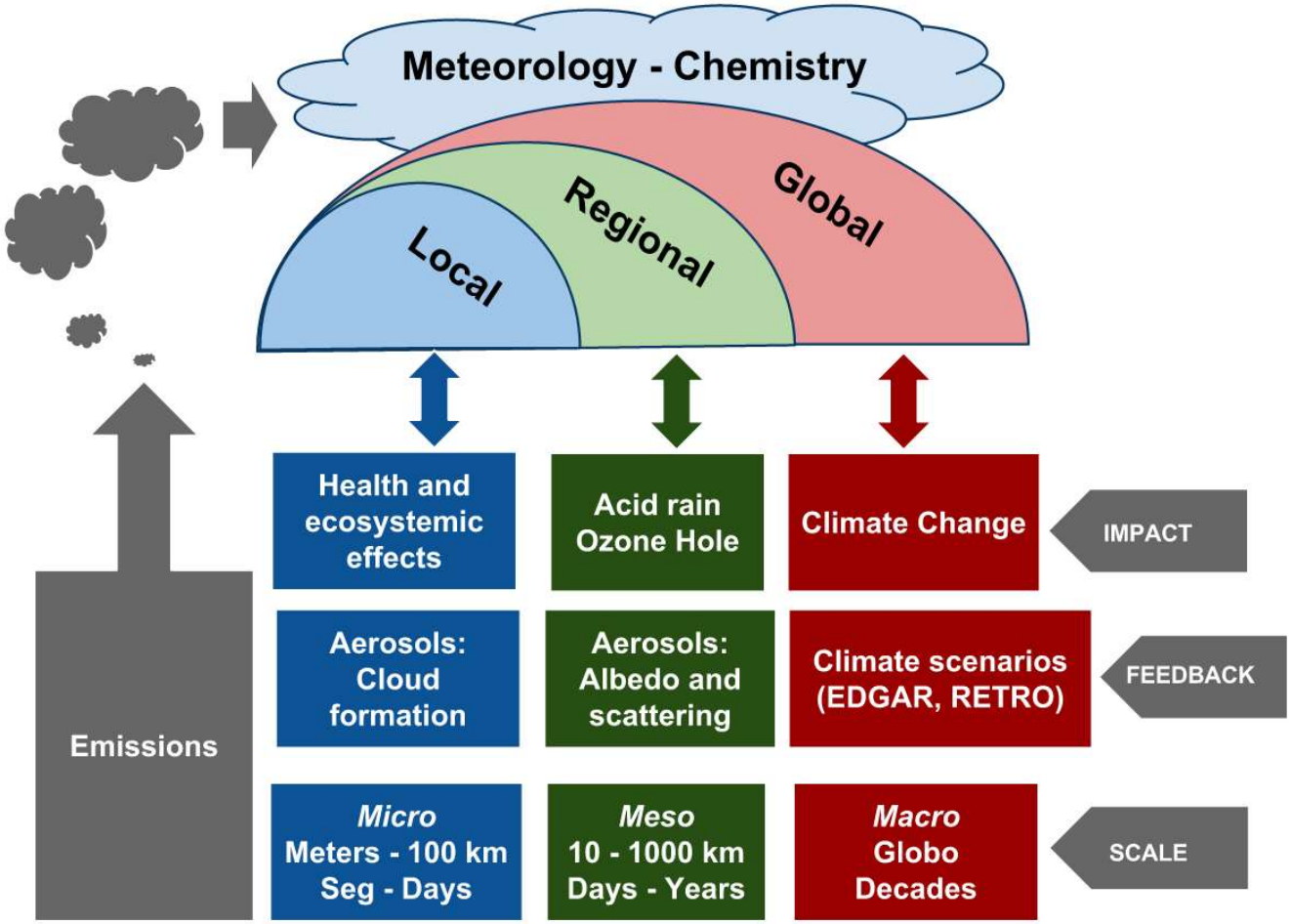
Motivation


Meteorology - Chemistry
Local
Regional
Global
Emissions
Health and ecosystemic effects
Acid rain Ozone Hole
Climate Change
IMPACT
Aerosols: Cloud formation
Aerosols: Albedo and scattering
Climate scenarios (EDGAR, RETRO)
FEEDBACK
Micro Meters - 100 km Seg - Days
Meso 10 - 1000 km Days - Years
Macro Globo Decades
SCALE

Sources of emissions inventories

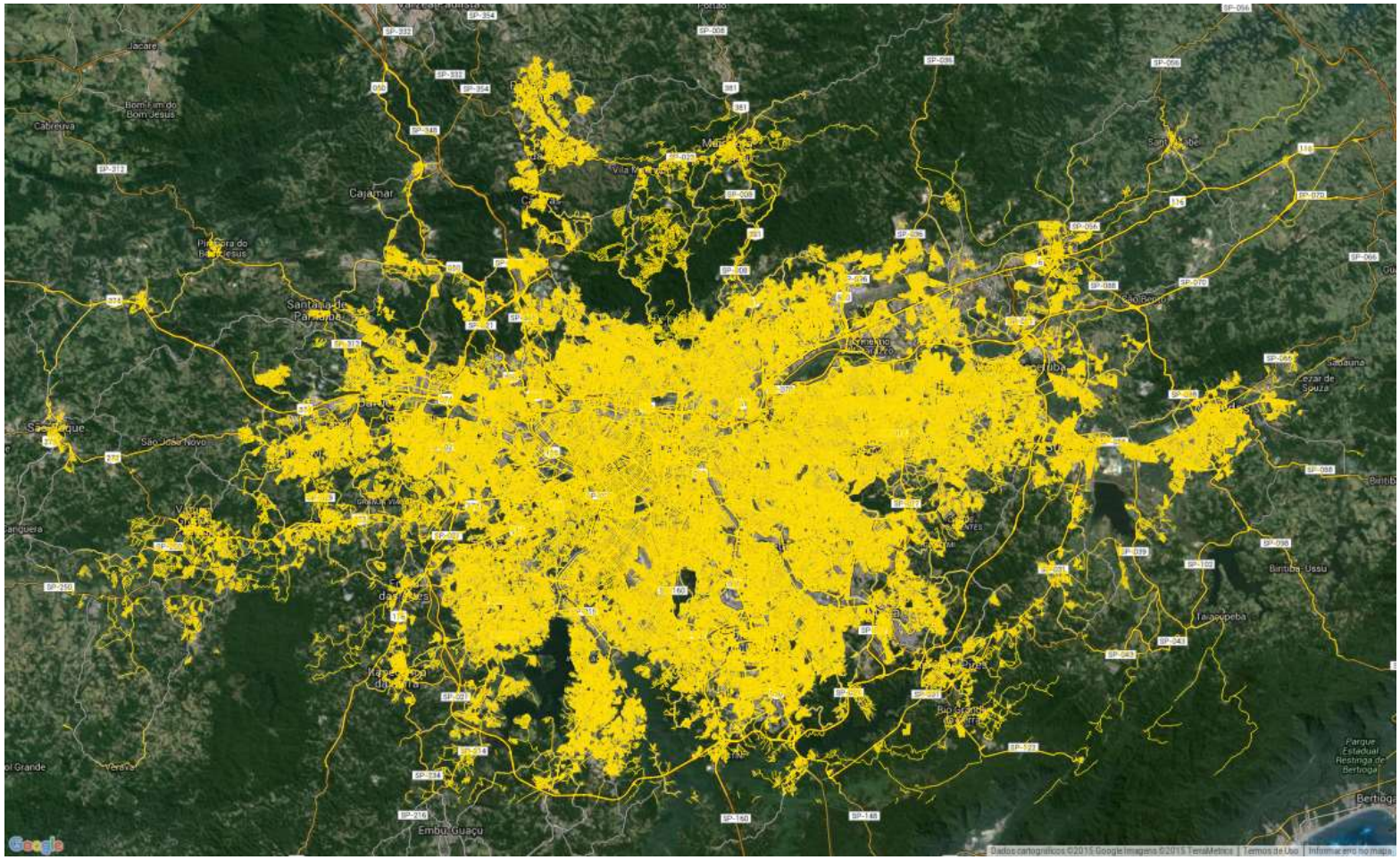
Vehicular emissions

1. Exhaust (Hot and cold)
2. Wear (Tyre and break)
3. Evaporatives
4. Fugitive dust
5. Refueling

Objectives

1. To develop a bottom-up (by street,hour and day) vehicular emissions inventory for São Paulo
2. Model emissions with WRF Chem

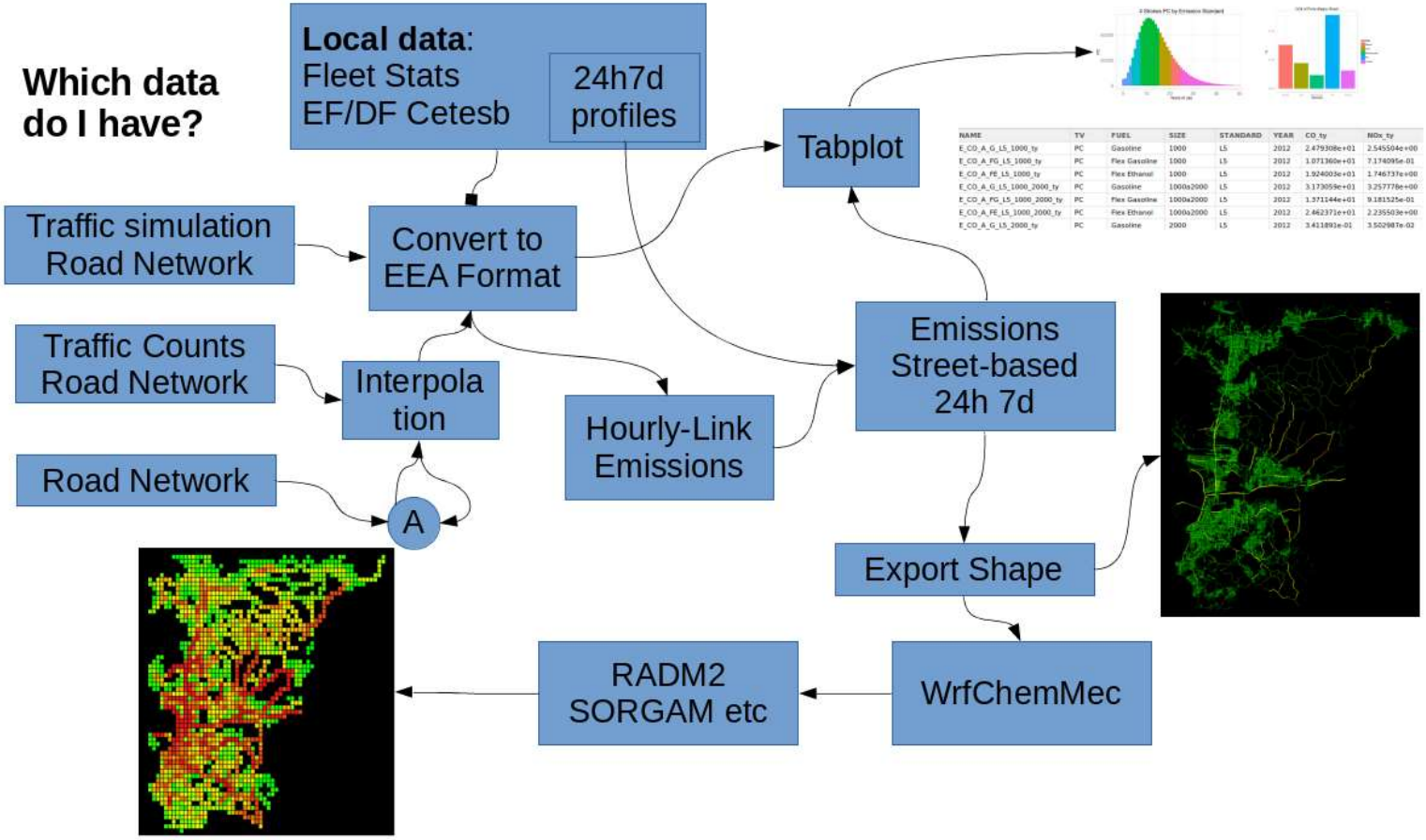
Road network



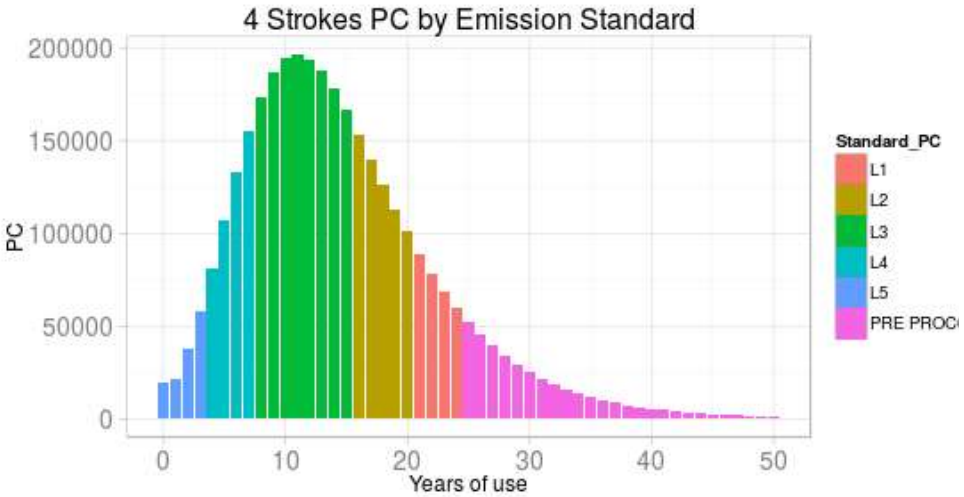
Material and methods

1. Road network from Open Street Map
2. remIAG_0.3.3
3. Statistics from Anfavea, CETESB, ANP, Maplink
4. R 3.2.2 ggplot2, ggmap, maptools, ncdf, rgeos

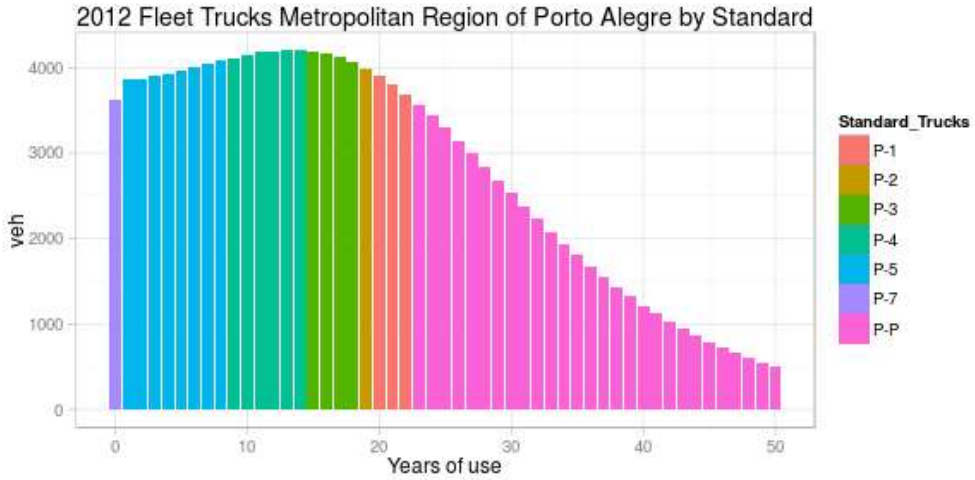
Method remIAG 0.3.3 (Ibarra, 2015)



PC- 14 years average age

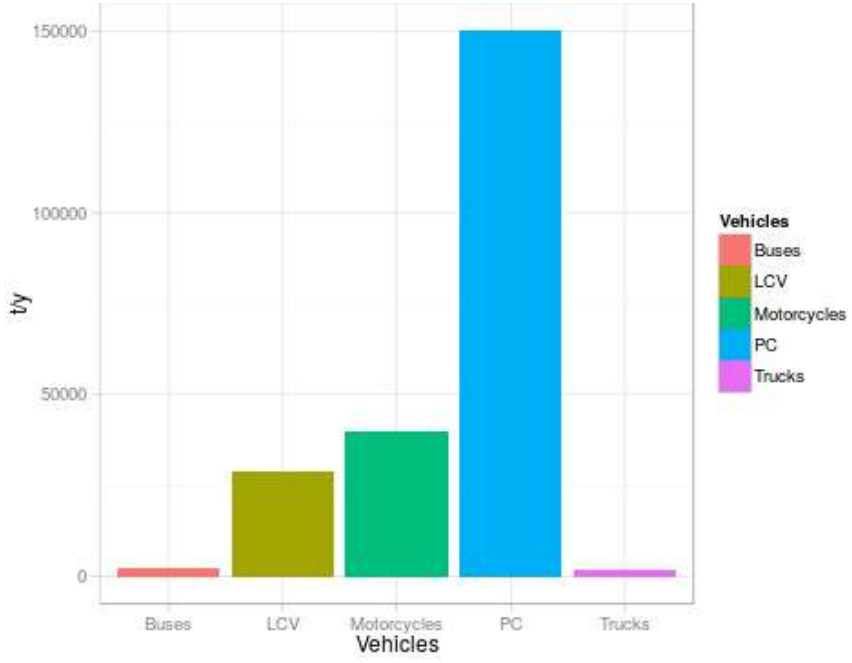


Trucks- 18 years average age



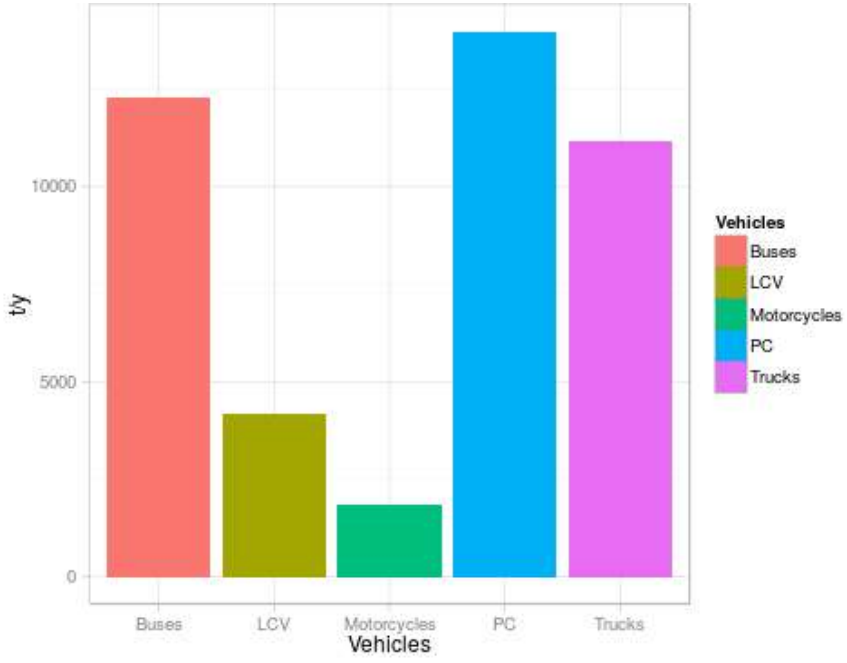
CO Emissions 2012 (t/y)

- 223,635.9 (t/y)



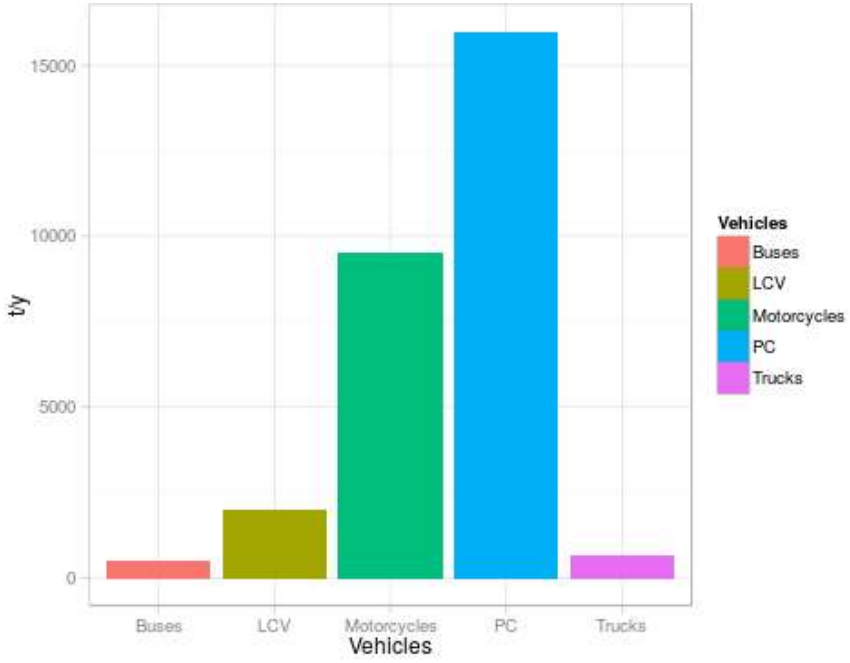
NOx Emissions 2012 (t/y)

- 43,432.97 (t/y)



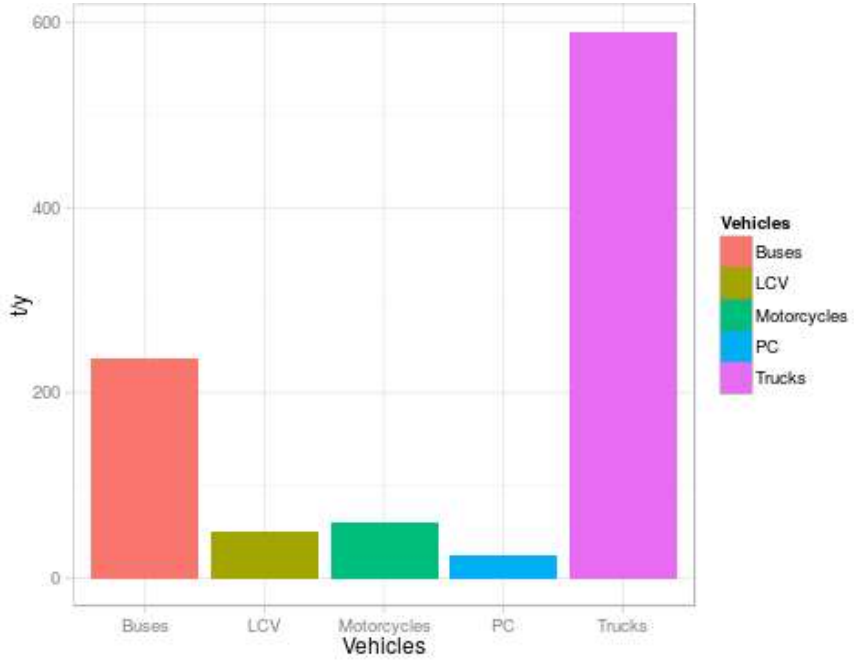
HC Emissions 2012 (t/y)

- 28,660.4 (t/y)



MP2.5 Emissions 2012 (t/y)

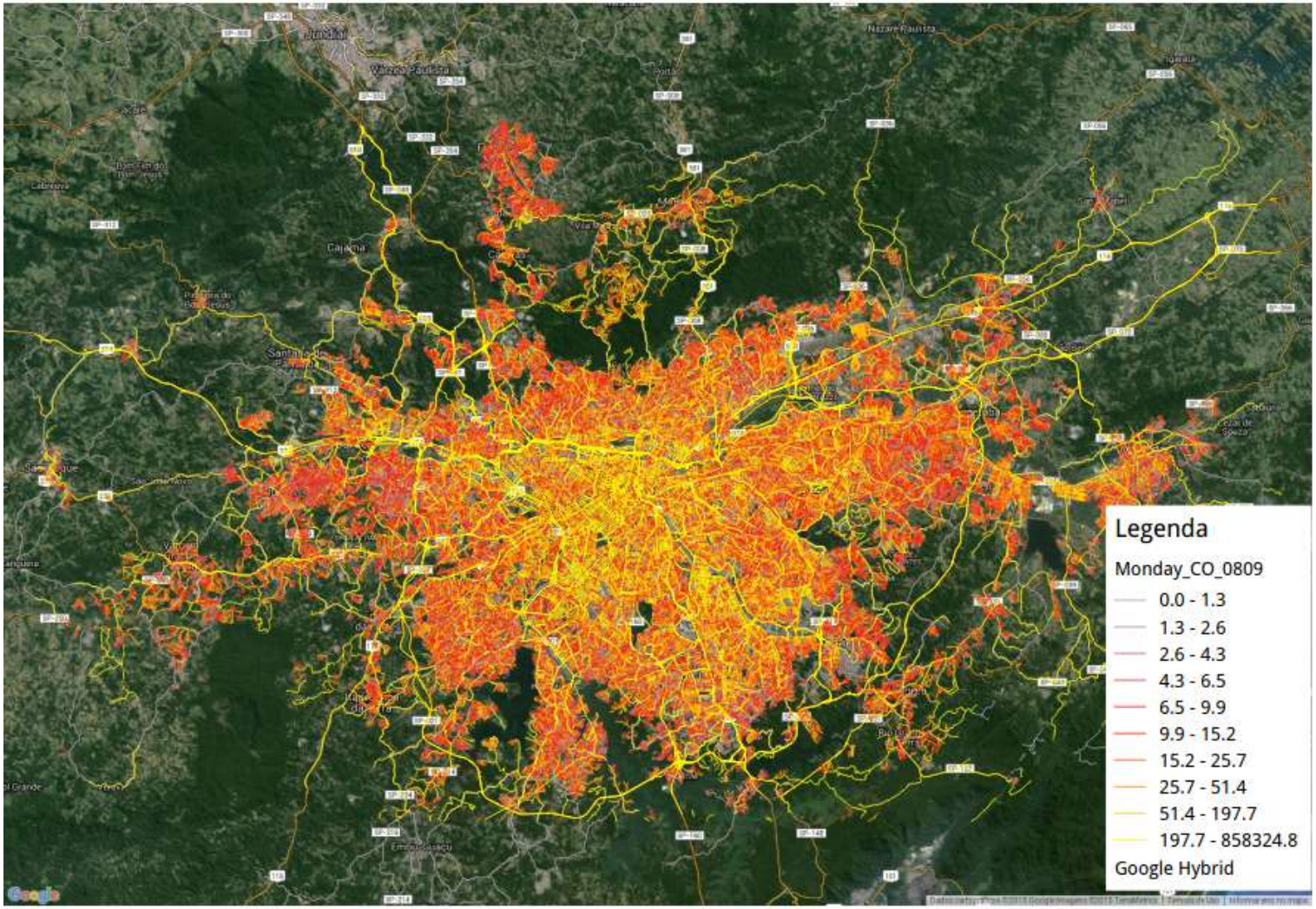
- 962.66 (t/y)



Emissions in São Paulo 2012 (t/y)

	TV	FUEL	CO	NOX	HC	MP2_5	FC	CO2	SO2
1	PC	Gasoline	88278	8093	9222	23	988950	2786488	396
2	PC	Flex	2282	367	403	1	240954	530344	97
3	PC	Ethanol	59815	5519	6363	0	1021832	713389	409
4	LCV	Gasoline	3676	289	273	2	118829	334816	48
5	LCV	Flex	215	18	32	0	48982	109556	20
6	LCV	Ethanol	24838	2175	1604	0	336612	643283	135
7	LCV	Diesel	346	1668	104	48	72752	228390	29
8	T-Light	Diesel	1331	7699	453	406	199168	625241	80
9	T-Medium	Diesel	280	1578	94	84	40095	125870	16
10	T-Heavy	Diesel	323	1892	107	99	48043	150819	20
11	Bus	Diesel	2339	12290	512	237	551007	1729759	220
12	Motorcycles	Gasoline	38072	1719	9216	55	245494	691709	98
13	Motorcycles	Flex	1840	126	276	6	46333	104455	19
14	Total	Total	223635	43433	28659	961	3959051	8774119	1587

Hourly emissions of CO (g/h)



Hourly emissions of CO (g/h)